AIA - Assignment - 2

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Q1 model has $d_e = 4$ & ~~position~~ token - & positional - embedding matrices.

$$W_e = \begin{pmatrix} 2 & 0 & 1 & 2 & 1 \\ 1 & 1 & 0 & 0 & 2 \\ 0 & 2 & 1 & 3 & 1 \\ 1 & 0 & 2 & 1 & 0 \end{pmatrix}, \quad W_p = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ 1 & 0 & 1 \\ 2 & 0 & 1 \end{bmatrix}$$

where vocabulary index runs from 1-5 & position from 1-3.

a) Examine how the input to the first layer is formed for token $v=4$ at position $t=2$: Identify the column of W_e & column of W_p used & determine the embedding $e = W_e[:, v] + W_p[:, t]$

given $v = 4$ & $t = 2$.

$W_e[:, 4] = \begin{bmatrix} 2 \\ 0 \\ 3 \\ 1 \end{bmatrix}$ // 4th column of W_e.

$W_p[:, 2] = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$ // 2nd column of W_p.

Final embedding:-

$$e = [W_e[:, v] + W_p[:, t]$$
$$= W_e[:, 4] + W_p[:, 2]$$
$$= \begin{bmatrix} 2 \\ 0 \\ 3 \\ 1 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 3 \\ 1 \end{bmatrix}$$

token embedding :- $\Rightarrow (2, 0, 3, 1)^T$

positional embedding :- $\rightarrow (1, 1, 0, 0)^T$

final input embedding :- $e = (3131)^T$

b) Apply the effect of Layer Norm, with $\gamma = 1$ & $\beta = 0$, on embedding e.

from part (a): determine the mean m, the variance $\sigma^2 = \frac{1}{d_e} \sum_i (e_i - m)^2$ & normalized vect $= \frac{(e-m)}{\sigma}$, and deduce the mean and variance of the normalize vector..

from (a) $\Rightarrow$ we have :- $e = \begin{bmatrix} 3 \\ 1 \\ 3 \\ 1 \end{bmatrix}$ $d_e = 4$, $\gamma = 1$, $\beta = 0$

→ mean ⇒ $m = \frac{1}{de} \sum_{i=1}^{de} e_i$

$\therefore \quad m = \frac{3+1+3+1}{4} = \frac{8}{4} = 2.$

→ variance ⇒ $\sigma^2 = \frac{1}{de} \sum_{i=1}^{de} (e_i - m)^2$

$\sigma^2 = \frac{(3-2)^2 + (1-2)^2 + (3-2)^2 + (1-2)^2}{4}$

$\sigma^2 = \frac{1^2 + (-1)^2 + (1)^2 + (-1)^2}{4} = \frac{4}{4} = 1.$

$\sigma^2 = 1.$

→ Standard deviation

$\sigma = \sqrt{\sigma^2} = \sqrt{1^2} = \underline{1}$

→ normalized vector

$\hat{e} = \frac{e-m}{\sigma} = \frac{1}{1} \begin{bmatrix} 3-2 \\ 1-2 \\ 3-2 \\ 1-2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$

give $\alpha = 1$, $\beta = 0$.

$CN(e) = \alpha \frac{e-m}{\sigma} + \beta.$

$= 1 \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix} + 0 = \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$

For Normalize vector

mean = $m' = \frac{(-1+1-1)_2}{n} = 0$

variance = $\sigma^{2'} = \frac{1^2 + (-1)^2 + 1^2 + (1-1^2)}{4} = \frac{1+1+1}{4} = \frac{3}{3} = 1$